

Rehaan Mirajwale

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CAREER PROFILE

AI Engineer with hands-on experience designing, evaluating, and deploying Machine Learning and Generative AI systems in production environments. Specialized in LLM pipelines, multi-agent workflows, Retrieval-Augmented Generation (RAG), real-time speech recognition (ASR), and model optimization, with a proven record of translating research into scalable, production-grade solutions using Python, PyTorch, FastAPI, and Docker.

TECHNICAL SKILLS

Programming & Tools: Python, C programming, SQL, Excel, Git, Docker, Kubernetes

Generative AI & Machine Learning: LLMs, RAG, Multi-Agent Systems, Diffusion Models, Deep Learning, LangChain, Machine Learning, NLP, Predictive Modeling, Prompt Engineering, Model Fine-Tuning

Frameworks & Libraries: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, Pandas, FastAPI

Data Science & Analytics: EDA, Statistical Modeling, Forecasting, Data Visualization, KPI Reporting

Cloud & Deployment: API-Based Model Integration, WebSocket Streaming, Containerized Deployment (Docker)

EXPERIENCE

1. Junior AI Engineer | Cynergy AI & Automation

February 2026 – May 2026

- Engineered a multilingual real-time ASR pipeline leveraging Faster-Whisper and ElevenLabs Speech-to-Text APIs, delivering streaming transcription, automatic language detection, and speaker diarization across 10+ Indian languages.
- Designed and implemented a transcript stitching and post-processing engine (Smart Stitcher) that resolved overlap artifacts and eliminated duplicate segments in rolling-window audio processing, significantly improving transcription quality and downstream usability.
- Developed an LLM-powered clinical information extraction pipeline generating structured diagnoses, symptoms, and recommendations from live medical conversations, enforcing transcript-grounded validation to minimize hallucinations and integrating outputs via FastAPI WebSockets for real-time delivery

EDUCATION

University of Liverpool, United Kingdom

September 2024 – September 2025

MSc – Data Science and Artificial Intelligence

Relevant Modules: Computational Intelligence, Data Mining, Applied Artificial Intelligence

KLS Gogte Institute of Technology, India

August 2020 – July 2024

B.E – Electronics and Communication Engineering

Relevant Modules: C programming, Digital Electronics, Signal Processing, Network analysis, VHDL

PROJECTS

1. Generative Diffusion Model for Kagome Lattice Crystal Structure Design [MSc]

- Fine-tuned Microsoft MatterGen with adapter-based learning to bias outputs toward rare Kagome structures addressing a severe class-imbalance challenge in materials discovery.
- Built evaluation pipeline (SPP, SOAP, AMD/PDD) achieving 2× higher discovery rate and 80% structural validity.
- Automated data preprocessing and validation, trimming compute cost by 30%

2. Local RAG-Based QA System [MSc]

- Built a fully offline Retrieval-Augmented Generation (RAG) system to improve LLM reliability by grounding responses strictly in user-provided documents, reducing hallucinations common in generative AI systems.
- Designed an end-to-end pipeline for document ingestion, chunking, embedding generation using Hugging Face transformer models, and semantic retrieval with a local ChromaDB vector database.
- Implemented a post-generation verification step to validate outputs against retrieved context, improving factual and numeric accuracy and demonstrating practical GenAI system evaluation and reliability techniques

3. Image-to-Video Multi-Agent Pipeline (LangGraph, RAG, Gemini)

- Architected a 7-node multi-agent LangGraph pipeline converting photo directories and text prompts into fully narrated videos, orchestrating Gemini vision-language analysis, storyboard generation, TTS narration, and programmatic rendering via Remotion.
- Designed a dual-collection RAG system (ChromaDB + sentence-transformers) with task-specific chunking strategies, grounding storyboard style and generated Remotion/React code in retrieved context.

CERTIFICATIONS

- Oracle – Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- NPTEL - The Joy of Computing Using Python
- NPTEL - English Language for Competitive Exams